

typed-dsa + Touying

Declarative data-structure diagrams directly inside presentation slides.

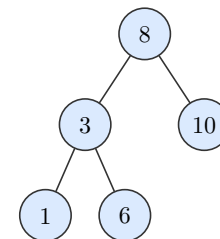
Same source, presentation-ready output

Typst source

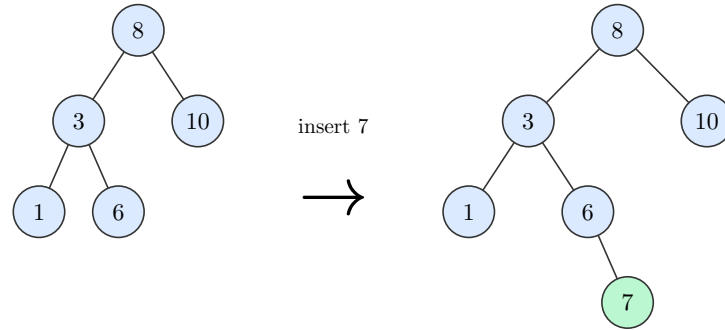
```
#bst(8, 3, 10, 1, 6).diagram
```

Keep examples editable in the same
file as the slides.

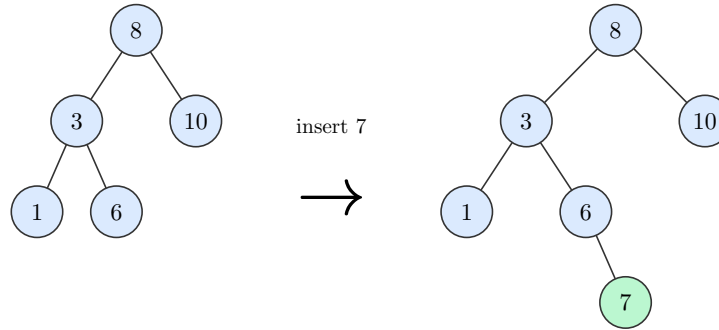
Rendered diagram



Explain operations step by step



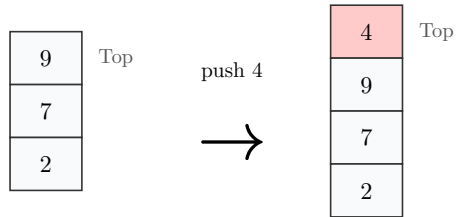
Explain operations step by step



The operation label, before/after state, and highlighted node are generated from the structure operation.

Linear structures fit lecture examples

Stack

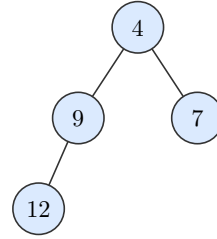


Queue

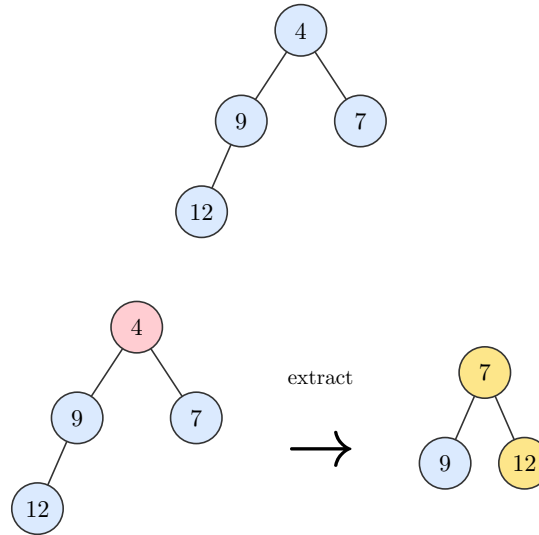
Front, Rear

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Animated reveal works normally



Animated reveal works normally



Use Touying reveals for the teaching flow; typed-dsa only handles the diagram.

Bubble sort, one step per reveal

start

5	1	4	2
0	1	2	3

Bubble sort, one step per reveal

compare 5 and 1

5	1	4	2
0	1	2	3

Bubble sort, one step per reveal

swap 5 and 1

1	5	4	2
0	1	2	3

Bubble sort, one step per reveal

compare 5 and 4

1	5	4	2
0	1	2	3

Bubble sort, one step per reveal

swap 5 and 4

1	4	5	2
0	1	2	3

Bubble sort, one step per reveal

compare 5 and 2

1	4	5	2
0	1	2	3

Bubble sort, one step per reveal

swap 5 and 2

1	4	2	5
0	1	2	3

Bubble sort, one step per reveal

compare 1 and 4

1	4	2	5
0	1	2	3

Bubble sort, one step per reveal

compare 4 and 2

1	4	2	5
0	1	2	3

Bubble sort, one step per reveal

swap 4 and 2

1	2	4	5
0	1	2	3

Bubble sort, one step per reveal

compare 1 and 2

1	2	4	5
0	1	2	3

Bubble sort, one step per reveal

sorted

1	2	4	5
0	1	2	3